

Tutorial-2.

2403A52202

A mobile service provider wants to classify sms message as spam or Not spam using a simple neural network (perceptron)

x_1 = specific words and phrases

x_2 = Links and Attachments

x_3 = Bias $\rightarrow$ constant

control the boundary of models

x_1	x_2	x_3	Target(y) (yes/no) question
4	2	1	1
1	0	1	0
6	3	1	1
2	1	1	0

① Inputs : x_1, x_2

② weights : w_1, w_2

③ Bias (b)

④ weighted sum : $z = x_1 w_1 + x_2 w_2 + b$

$$= \sum_{i=1}^n x_i w_i + b$$

⑤ Activation function (Here : step function)

$$y^n = \begin{cases} 0 & ; z < 0 \\ 1 & ; z \geq 0 \end{cases}$$

⑥ Updated weights

$$w_{i\text{new}} = w_{i\text{old}} + \eta (y - y^n) x_i$$

η = Learning rate = ①

y = actual target / True target

y^n = predicted output / Target

$$x_1 = 4, x_2 = 2, b = 1$$

$$\text{Let } w_1 = 0, w_2 = 0 \quad \eta = 1$$

Inputs =

$$z = ?$$

$$z = w_1 x_1 + w_2 x_2 + b$$

$$z = 0(4) + (0)(2) + 1$$

$$z = 1 \quad \text{so check } y^n = 1$$

The y^n is matched with target (y) in 1st row

Iteration = epoch $\rightarrow 10$

③ An email service provider wants to classify incoming emails as spam or not spam, using a single network model.

The training dataset is shown below

x_1	x_2	x_3	Target (y)
5	3	1	0
1	0	1	0
7	5	1	1
2	1	1	0
6	4	1	1

① Inputs

$$x_1 = 5, x_2 = 3, b = 1$$

$$\text{Let } w_1 = 0, w_2 = 0, \eta = 1$$

$$z = w_1 x_1 + w_2 x_2 + b$$

$$z = (0)(5) + (0)(3) + 1$$

$$z = 1 \quad \text{so } y^n = 1$$

$$w_{1\text{new}} = w_{1\text{old}} + \eta (y - y^n) x_1$$

$$\begin{aligned} w_{1\text{new}} &= (0) + 1 (0 - 1) 5 \\ &= 0 + (-5) \end{aligned}$$

$$w_{1\text{new}} = -5$$

$$w_{2\text{new}} = w_{2\text{old}} + \eta (y - y^n) x_2$$

$$= 0 + (0 - 1) 3$$

$$w_{2\text{new}} = -3$$

$$w_1 = -5, w_2 = -3, \eta = 1, \gamma = 1, \gamma_2 = 0, b = 1$$

$$z = w_1 x_1 + w_2 x_2 + b$$

$$= (-5)(1) + (-3)(0) + 1$$

$$= -5 + 1$$

$$\boxed{z = -4} \quad \boxed{\hat{y} = 0} \rightarrow \text{The output of old is predicted.}$$

$$w_{\text{new}} = (-5) + 1(y - \hat{y})x_1$$

$$= (-5) + 1(0 - 0)1$$

$$\boxed{w_{\text{new}} = -5}$$

If we update weights after output is correct then the same weights will come.

An e-commerce company predicts whether a customer will make a purchase based on time spent on website (minutes) and number of items viewed.

Dataset :

Time on site (x_1)	Items viewed (x_2)	Bias (x_3)	Target (purchase = $\frac{1}{N_0} = 0$)
2	3	1	0
15	25	1	1
8	12	1	1
3	5	1	0
12	20	1	1

① Inputs :

$$x_1 = 2, x_2 = 3, b = 1, w_1 = 0, w_2 = 0, \eta = 1$$

$$z = w_1 x_1 + w_2 x_2 + b$$

$$= (0)(2) + (0)(3) + 1$$

$$\boxed{\hat{y} = 1}$$

$$w_{1, \text{new}} = w_{1, \text{old}} + \eta (y - y^{\wedge}) x_1$$

$$w_1 = 0 + 1(0-1)2$$

$$\boxed{w_1 = -2}$$

$$w_{2, \text{new}} = 0 + 1(0-1)3$$

$$w_2 = -3$$

2nd case

$$w_1 = -2, w_2 = -3, b = 1, x_1 = 15, x_2 = 25, \eta = 1$$

$$z = w_1 x_1 + w_2 x_2 + b$$

$$= (-2)(15) + (-3)(25) + 1$$

$$= -30 + -75 + 1$$

$$= -104$$

$$\boxed{y^{\wedge} = 0}$$

$$w_{1, \text{new}} = w_{1, \text{old}} + \eta (y - y^{\wedge}) x_1$$

$$w_{1, \text{new}} = -2 + 1(1-0)15$$

$$= -2 + 15$$

$$\boxed{w_1 = 13}$$

$$w_{2, \text{new}} = w_{2, \text{old}} + \eta (y - y^{\wedge}) x_2$$

$$= -3 + 1(1-0)25$$

$$= -3 + 25$$

$$\boxed{w_2 = 22}$$

$$w_1 = 13, w_2 = 22, \eta = 1$$

$$x_1 = 8, x_2 = 12, b = 1$$

$$z = w_1 x_1 + w_2 x_2 + b$$

$$z = (13)(8) + (22)(12) + 1$$

$$z = 104 + 246 + 1$$

$$\boxed{z = 351}$$

$$\boxed{y^{\wedge} = 1}$$

At the third row we have predicted

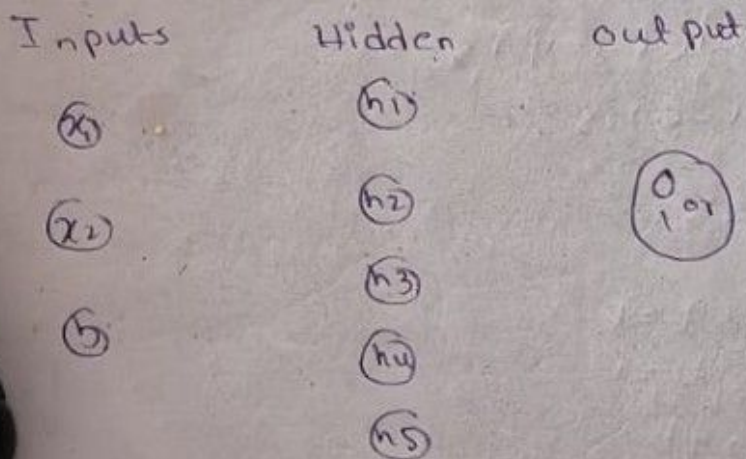
An email service provider wants to classify incoming emails as spam or not spam using a single layer network model with 5 hidden neurons. The training dataset is shown below.

x_1	x_2	x_3 (b)	Target (Y)
5	3	1	1
1	0	1	0
7	5	1	1
2	1	1	0
6	4	1	1

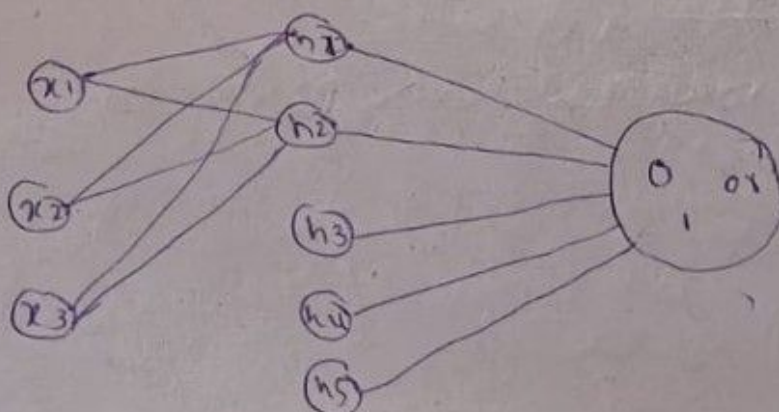
$$b = w_0 = x_3$$

Hidden layers are which is located like a mediator b/w inputs and outputs.

Single layer neural network.



Everything is works / performed by hidden neuron. Each hidden neuron is connected to all the inputs (x_1, x_2, b) in a single input layer.

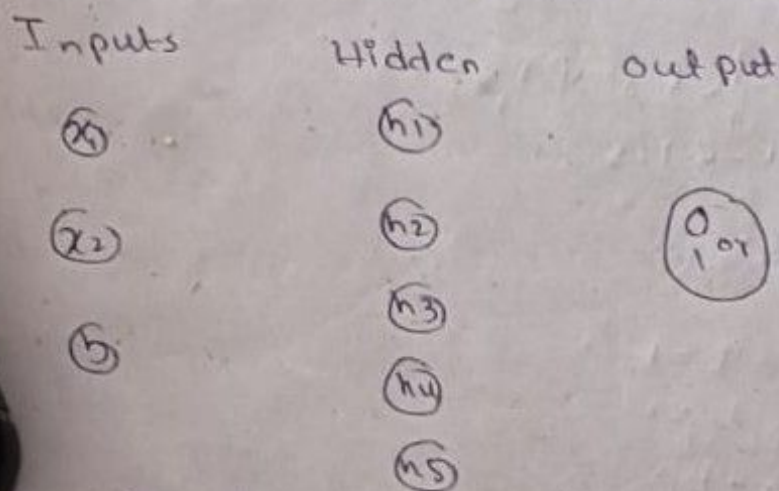


An email service provider wants to classify incoming emails as spam or not spam using a single layer network model with 5 hidden neurons. The training dataset is shown below.

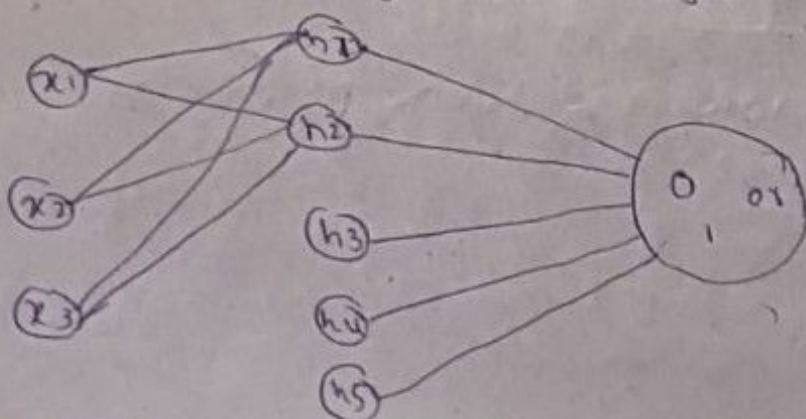
x_1	x_2	x_3	(b) Target (Y)	$b = w_0 = x_3$
5	3	1	1	
1	0	1	0	
7	5	1	1	
2	1	1	0	
6	4	1	1	

Hidden layers are which is located like a mediator b/w inputs and outputs.

single layer neural network



Everything is works / performed by hidden neuron. Each hidden neuron is connected to all the inputs (x_1, x_2, b) in a single input layer.



	w_1	w_2	b
h_1	0	0	1
h_2	0.1	-0.2	1
h_3	0.3	0.1	1
h_4	0.2	-0.2	1
h_5	0.4	0.1	1

$$z_1 = w_1 x_1 + w_2 x_2 + b$$

$$= (0) + (0) + 1$$

$$= 1$$

$h_2 \Rightarrow$

$$z_2 = w_1 x_1 + w_2 x_2 + b$$

$$z = (0.1)(5) + (-0.2)(3) + 1$$

$$= 0.9$$

$h_3 \Rightarrow$

$$z_3 = (0.3)(5) + (0.1)(3) + 1$$

$$= 2.8$$

$h_4 \Rightarrow$

$$z_4 = (0.2)(5) + (-0.2)(3) + 1$$

$$= 1.4$$

$h_5 \Rightarrow$

$$z_5 = (0.4)(5) + (0.1)(3) + 1$$

$$= 3.3$$

$$h_1 = \sigma(z_1) = \frac{1}{1 + e^{-z_1}} = \frac{1}{1 + e^{-1}} = 0.74$$

$$h_2 = \sigma(z_2) = \frac{1}{1 + e^{-z_2}} = \frac{1}{1 + e^{-0.9}} = 0.72$$

$$h_3 = \sigma(z_3) = \frac{1}{1 + e^{-z_3}} = \frac{1}{1 + e^{-2.8}} = 0.95$$

$$h_4 = \sigma(z_4) = \frac{1}{1 + e^{(-z_4)}} = \frac{1}{1 + e^{(-1.4)}} = 0.80$$

$$h_5 = \sigma(z_5) = \frac{1}{1 + e^{(-z_5)}} = \frac{1}{1 + e^{(-3.8)}} = 0.97$$

$$z_{out} = h_1 v_1 + h_2 v_2 + h_3 v_3 + h_4 v_4 + h_5 v_5$$

$$\sum_{i=1}^N h_i v_i$$

$h \rightarrow$ hidden layers and inputs

$v \rightarrow$ hidden layers and outputs

$v = 0.1, 0.2, 0.3, 0.4, 0.5, -0.1, -0.5$ any value

$$\sigma(z_{out}) = ?$$

$$z_{out} = (0.74)(0.1) + (0.72)(0.2) + (0.95)(0.3) + (0.80)(-0.2) + 0.97(0.5) = 0$$

$$\sigma(z_{out}) = \frac{1}{1 + e^{(-z_{out})}} = \frac{1}{1 + e^{(-0)}} = 0.5$$

$$\sigma(z_{out}) \Rightarrow 0.999 \approx 1.0$$

$$\sigma(z_{out}) > 0.5 \quad \boxed{y^* = 1}$$

if $z_{out} \geq 0.5 \Rightarrow 1$
 $< 0.5 \Rightarrow 0$ } mission will detect.

$$w_{i,new} = w_{i,old} + \eta (y - \hat{y}) x_i$$

$$v_{i,new} = v_{i,old} + \eta (y - \hat{y}) h_i$$

If the output is less than 0.5 then we want to update the w, v values both/ any one of it

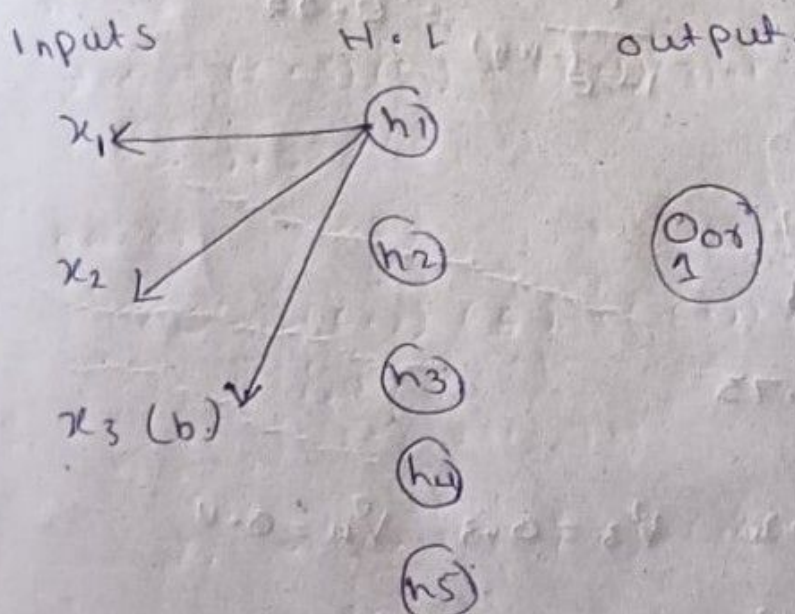
A mobile service provider wants to classify sms message as spam or not spam using a single-layer neural network with 5 hidden neurons in the layer.

x_1 = specific words and phrases

x_2 = links and attachments

x_3 = bias

x_1	x_2	x_3	Target (y)
4	2	1	1
1	0	1	0
6	3	1	1
2	1	1	0



h_1

$$z_1 = w_1 x_1 + w_2 x_2 + b$$

$$z = (0.17)(4) + (0.2)(2) + 1$$

$$= 1.8$$

h2

$$z_2 = w_1 x_1 + w_2 x_2 + b$$

$$= (-0.2)(4) + (0.3)(2) + 1$$

$$= -0.8$$

h3

$$z_3 = w_1 x_1 + w_2 x_2 + b$$

$$= (0.4)(4) + (-0.1)(2) + 1$$

$$= 2.4$$

h4

$$z_4 = w_1 x_1 + w_2 x_2 + b$$

$$= (0.3)(4) + (0.2)(2) + 1$$

$$= 2.6$$

h5

$$z_5 = w_1 x_1 + w_2 x_2 + b$$

$$= (0.2)(4) + (-0.4)(2) + 1$$

$$= 1$$

$$h_1 = \sigma(z_1) = \frac{1}{1 + e^{-z_1}} = \frac{1}{1 + e^{-1.8}} = 0.86$$

$$h_2 = \sigma(z_2) = 0.69$$

$$h_3 = \sigma(z_3) = 0.92$$

$$h_4 = \sigma(z_4) = 0.93$$

$$h_5 = \sigma(z_5) = 0.73$$

$$\vartheta = 0.1, \quad \vartheta_2 = -0.2, \quad \vartheta_3 = 0.3, \quad \vartheta_4 = 0.4$$

$$\vartheta_5 = -0.3$$

$$z_{out} = h_1 \vartheta_1 + h_2 \vartheta_2 + h_3 \vartheta_3 + h_4 \vartheta_4 + h_5 \vartheta_5$$

$$= 0.86(0.1) + 0.69(-0.2) + (0.92)(0.3)$$

$$+ 0.93(0.4) + 0.73(-0.3)$$

$$= 0.38$$

$$\sigma(\text{net}) = \frac{1}{1 + e^{-(\text{net})}}$$

$$\sigma(\text{net}) = 0.5 \text{ or } 0.6$$

$$\boxed{\begin{matrix} \sigma \geq 0.5 \rightarrow \hat{y} = 0 \\ \sigma < 0.5 \rightarrow \hat{y} = 1 \end{matrix}} \quad \text{if } \sigma(\text{net}) \geq 0.5$$

$$\text{So } \boxed{\hat{y} = 1}$$

$\hat{y}$ is y predicted

$$\text{Error} = \text{Actual} - \underset{\substack{\uparrow \\ \text{Learning} \\ \text{Rate}}}{\eta} (\underset{\substack{\uparrow \\ \text{true}}}{y} - \underset{\substack{\uparrow \\ \text{predicted}}}{\hat{y}}) h_i$$

$$\boxed{w_{old} = w_{old} + \eta (y - \hat{y}) x_i}$$

we can update both v & w (or) any one of it, if output not predicted.